جامعة عفت
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Report 2:

Literature Review

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CASSIA: A Constraint-Based Intelligent Agent for Coordinating Solar, Waste-Heat, Cooling, and Water Sourcing in AI Data Centers

1. Introduction

In 2024, all data centers worldwide used about 415 TWh of electricity, about 1.5% of the world total. In its Base Case, the IEA projects about 945 TWh by 2030, with AI as the largest driver [1]. Li et al. project that global AI may withdraw 4.2–6.6 billion m³ of water in 2027; withdrawal is water taken from sources, and only part of it is consumed. They also show that saving carbon does not always save water [2].

Our project, CASSIA, is an agent that plans every hour. It shares solar power and waste heat between AI jobs, cooling, and alternative water production by membrane distillation (MD) or forward osmosis (FO). The goal is less grid power and city water, with all jobs on time, servers within safe temperatures, and zero thermal-throttling events for high-priority jobs.

We searched the ACM Digital Library, IEEE Xplore, ScienceDirect, and Google Scholar (2020–2026), combining "data center" or "scheduling" with terms such as "water," "cooling," "constraint programming," "A*," and "MDP." We kept peer-reviewed papers with a tested method. We review eight studies [3]–[10], grouped by method. Two come from other fields ([5] factories, [6] EV charging); we reuse only their problem structure. Section 4 adds two integrated works [11], [12]; ML-based [7] and [11] are comparators only, since CASSIA's core is classical AI.

2. Related Work

2.1 Scheduling jobs with constraints

Schweisgut et al. [3] schedule tasks on a cluster with a changing green power budget, testing 34 workflows of up to 30,000 tasks. With a deadline, the problem is easy on one processor (dynamic programming) but NP-hard on many. So they use greedy rules plus hill-climbing local search, and compare with an exact ILP only up to about 200 tasks, because the solver becomes too slow.

Jiang et al. [4] built WaterWise. It places batch jobs in data centers in different regions using mixed-integer linear programming (MILP) that weighs carbon and water. Delay limits are soft: if no plan meets them, it pays a penalty.

Terbrack and Claus [5] use constraint programming (CP) to schedule factory jobs under changing electricity prices. Due dates and setup times are constraints. On 20 test cases, energy cost fell by 5.3%, peak demand by 11.8%, and emissions by 8.3% on average. The goals conflict, so a person must rank them before solving.

None of the three controls cooling: [3] and [5] ignore water, and [4] models water but cannot change how the building is cooled.

2.2 Heuristic search

Shohan et al. [6] plan electric vehicle charging in buildings with solar power using A* search. The cost so far, g, is the energy cost at the real-time price, and the estimate h is forecast need times forecast price. The authors call h admissible but do not prove it; with solar and vehicle-to-grid sales, the remaining cost can fall below need × price, so optimality is unproven.

Qi et al. [7] built SHIELD for 16 geographically distributed data centers. It combines local search with an evolutionary algorithm, and a machine learning model picks where the local search starts. It returns trade-off plans for energy cost, carbon, and water footprint.

2.3 Rules and MDPs for facility control

Borkowski and Piłat [8] built a rule-based controller that decides when chillers switch between free cooling and compressor cooling. Its rules come from past plant data and were validated on an operating cooling system through 2017.

Bayati [9] models server power control as a Markov decision process (MDP). Value iteration finds the best policy, tested with Google job traces; the largest case has 103 servers. Neither [8] nor [9] uses solar power or water, or links cooling to job timing.

2.4 Using waste heat to make water

Yang et al. [10] co-plan a data center and a reverse osmosis (RO) plant less than 1 km apart in Jubail, Saudi Arabia. They schedule data center tasks, and waste heat warms the seawater, which changes RO

efficiency. They solve a two-stage stochastic MILP by Benders decomposition. Total yearly cost fell 86.5% against conventional designs, 17% of it from heat reuse. But the heat only preheats RO, which runs on electricity; it does not drive MD, FO, or storage, and its summary reports no cooling-mode choice. No peer-reviewed study we found links data center waste heat to FO.

3. Critical Comparison

Table 1 summarizes the studies. Datasets and baselines differ, so percentages show direction only. Only [3] checks against an exact optimum, and only [8] is a real-system test; the rest are simulations or planning studies.

Table 1. Summary of the reviewed studies and the two closest integrated works.

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
[3]	Greedy + local search; ILP check	Carbon-aware scheduling with a deadline; 34 workflows, up to 30,000 tasks	Median about 0.6 of ASAP baseline; about 0.15 for one heuristic at 3× deadline	Fixed task mapping; heuristics not optimal
[4]	MILP weighing carbon and water; soft delay limits	Job placement across regions; Google and Alibaba traces	About 21% less carbon, 14% less water vs. an unaware scheduler	Weak baseline; cooling not controlled
[5]	Constraint programming (CP)	Factory job shop with due dates and setup times; 20 instances	Energy cost −5.3%, peak −11.8%, emissions −8.3% (average)	Goals ranked by hand in a fixed order
[6]	A* search; state = (time step, battery charge)	EV charging with solar and vehicle-to-grid; 15-min steps, 24-h horizon	3–5% lower cost than particle swarm and genetic algorithms	Admissibility of h not shown; heuristic baselines only
[7]	Evolutionary + local search (ML-guided)	16 distributed data centers; 3 goals; hourly	2.1× hypervolume of next-best; up to 3.7× less carbon, 1.8× less water	Learned; no optimality proof; cooling not controlled
[8]	Rule-based controller	Chiller mode switching; operating data center, 2017	+2,064 h/year of free cooling; COP +69%; 206.9 MWh less electricity	Tuned to one plant
[9]	MDP solved by value iteration	Turning servers on/off; Google trace; largest case 103 servers	Optimal on/off policy and its structure	State space grows fast; identical servers and job lengths
[10]	Stochastic MILP; Benders decomposition	Co-planning a data center (tasks) and an RO plant; Jubail case	Yearly cost −86.5% vs. conventional; 17% from heat reuse	Heat only preheats RO; no MD/FO, storage, or cooling choice
[11]	Deep reinforcement learning (ML)	Jobs + cooling airflow; trace-based simulation	Up to 15% less energy than siloed and 10% less than joint baselines	Learned policy, not traceable; no solar, heat reuse, or water
[12]	Design-space search + carbon-aware scheduling	Sizing solar, wind, batteries; production traces	Combined scenario: 100% renewable coverage carbon-optimal in five regions	Design-time only; no cooling control or water

Optimality vs. speed. Exact methods are optimal for their model but grow fast: if n jobs each pick one of T start slots, there are up to T^n schedules. So [3] uses its ILP only up to about 200 tasks, while its heuristics handle 30,000 with no guarantee, and the MDP in [9] grows with servers × queue × lost jobs.

Hard vs. relaxed constraints. CP [5] keeps due dates hard but ranks its goals in a fixed order chosen beforehand, so under a cost-first order a slightly costlier plan with a much lower peak is never chosen.

WaterWise [4] relaxes delay limits with a penalty when no plan fits, which keeps a plan feasible but allows late jobs.

Deterministic vs. uncertain. Only [9] (job arrivals) and [6] (EV times) model uncertainty; [6] uses point price forecasts, and no study tests solar forecast errors.

Inspectable vs. learned. In rules [8], CP [5], MILP [4], and A* [6], each decision can be traced to a rule, constraint, or cost. SHIELD [7] reached the next-best method's plan quality with 34.4× less computing time, which helps with many sites and goals, but learned guidance is harder to trace.

Closest multi-resource methods, compared on the same six dimensions:

- **WaterWise [4]** — Objective: weighted carbon and water footprint. Constraints: soft delay limits. Scheduling: region and start time. Cooling: not controlled. Water: modelled, not produced. Limitation: weak baseline; no heat reuse.
- **SHIELD [7]** — Objective: cost, carbon, and water trade-offs. Constraints: no hard deadlines. Scheduling: hourly work across sites. Cooling: modelled, not controlled. Water: modelled, not produced. Limitation: learned guidance, no optimality proof.
- **DeepEE [11]** — Objective: total energy. Constraints: no overheating, balanced load. Scheduling: job placement. Cooling: airflow controlled. Water: not modelled. Limitation: learned policy, not traceable; no solar or heat reuse.

In all three, server heat is waste: none recovers it or trades cooling against water production. For CASSIA, WaterWise's soft limits would allow late high-priority jobs, and DeepEE's learned cooling choices could not be audited.

Threats to validity. Most results come from simplified simulations, and [8] and [10] each cover one site, so the gains may not carry over to a large AI data center.

4. Research Gap and Position of CASSIA

In the studies reviewed, we see two gaps.

- **No joint plan for jobs, power, cooling, and heat-to-water at one site.** Yang et al. [10] link job scheduling to water, but heat only preheats RO in a separate plant. DeepEE [11] couples jobs and cooling airflow but has no solar, heat reuse, or water. Carbon Explorer [12] is a design-time study with no cooling control or water. We did not identify an approach that schedules jobs, solar power, and cooling while routing waste heat among water production, heating, and storage.
- **No inspectable, multi-stage decision pipeline.** We call a method inspectable if every decision can be traced to a named rule, constraint, or cost. Rules [8], CP [5], and MILP [4], [10] meet this but each covers one or two stages; SHIELD [7] and DeepEE [11] cover more resources but rely on learned components.

Why combining existing tools is not enough. The stages form a chain: job scheduling → heat generation → cooling → heat recovery → water production. Each choice limits the next stage, so separate tools, even used together, miss these links and may conflict. CASSIA therefore needs a linked pipeline.

With one site, hourly decisions, hard deadlines, and safety-critical cooling, CASSIA (Report 1) uses exact, inspectable methods in a linked pipeline, re-planned hourly with new forecasts. Rules inspired by [8] pick the cooling mode (free, evaporative, or chiller), and new CASSIA rules pick the water source. A CSP then schedules jobs and cooling and assigns solar power, with deadlines, thermal limits, and zero throttling for high-priority jobs as hard constraints, and grid power and city water as weighted costs, avoiding the fixed goal ranking of [5]. A* then allocates the spare heat over 24 hours to heating loops, MD/FO water production, or thermal storage, on states (hour, storage level). As [6] shows, A* is optimal only if its heuristic never overestimates; CASSIA must prove this for its own model, and even then A* is optimal only for the heat plan given the CSP output. These are proposed design choices; Report 3 will test CASSIA against a baseline on grid electricity, municipal water, deadline compliance, and runtime.

References

[1] International Energy Agency, “Energy and AI,” IEA, Paris, France, Rep., Apr. 2025. Accessed: Nov. 10, 2026. [Online]. Available: https://www.iea.org/reports/energy-and-ai

[2] P. Li, J. Yang, M. A. Islam, and S. Ren, “Making AI less ‘thirsty’: Uncovering and addressing the secret water footprint of AI models,” *Commun. ACM*, vol. 68, no. 7, pp. 54–61, Jul. 2025, doi: 10.1145/3724499.

[3] D. Schweisgut, A. Benoit, Y. Robert, and H. Meyerhenke, “Carbon-aware workflow scheduling with fixed mapping and deadline constraint,” in *Proc. 54th Int. Conf. Parallel Process. (ICPP ’25)*, Sep. 2025, pp. 627–637, doi: 10.1145/3754598.3754673.

[4] Y. Jiang, R. Basu Roy, R. Kanakagiri, and D. Tiwari, “WaterWise: Co-optimizing carbon- and water-footprint toward environmentally sustainable cloud computing,” in *Proc. 30th ACM SIGPLAN Annu. Symp. Princ. Pract. Parallel Program. (PPoPP ’25)*, Mar. 2025, pp. 297–311, doi: 10.1145/3710848.3710891.

[5] H. Terbrack and T. Claus, “The generalized energy-aware flexible job shop scheduling model: A constraint programming approach,” *Comput. Ind. Eng.*, vol. 204, Art. no. 111065, Jun. 2025, doi: 10.1016/j.cie.2025.111065.

[6] M. J. A. Shohan, M. M. Islam, S. Owais, and M. O. Faruque, “Optimal energy management of EVs at workplaces and residential buildings using heuristic graph-search algorithm,” *Energies*, vol. 17, no. 21, Art. no. 5278, Oct. 2024, doi: 10.3390/en17215278.

[7] S. Qi, D. Milojicic, C. Bash, and S. Pasricha, “SHIELD: Sustainable hybrid evolutionary learning framework for carbon, wastewater, and energy-aware data center management,” in *Proc. 14th Int. Green Sustain. Comput. Conf. (IGSC ’23)*, Oct. 2023, pp. 56–62, doi: 10.1145/3634769.3634810.

[8] M. Borkowski and A. K. Piłat, “Energy efficiency increase achieved by dedicated rule-based control of chillers operating in the data center,” *Energies*, vol. 15, no. 7, Art. no. 2419, Mar. 2022, doi: 10.3390/en15072419.

[9] L. Bayati, “Discrete-time MDP policy for energy-aware data center,” in *Proc. 12th Int. Conf. Smart Cities Green ICT Syst. (SMARTGREENS)*, vol. 1, Apr. 2023, pp. 89–97, doi: 10.5220/0011846300003491.

[10] R. Yang, C. Konstantinou, and Y. Hou, “Enabling grid-interactive data center-desalination coordination through thermo-electric coupling-based waste heat utilization,” *Appl. Energy*, vol. 408, Art. no. 127428, Apr. 2026, doi: 10.1016/j.apenergy.2026.127428.

[11] Y. Ran, H. Hu, Y. Wen, and X. Zhou, “Optimizing energy efficiency for data center via parameterized deep reinforcement learning,” *IEEE Trans. Serv. Comput.*, vol. 16, no. 2, pp. 1310–1323, Mar.–Apr. 2023, doi: 10.1109/TSC.2022.3184835.

[12] B. Acun et al., “Carbon Explorer: A holistic framework for designing carbon aware datacenters,” in *Proc. 28th ACM Int. Conf. Archit. Support Program. Lang. Oper. Syst. (ASPLOS ’23)*, vol. 2, Mar. 2023, pp. 118–132, doi: 10.1145/3575693.3575754.

Appendix A — Verification Log

This appendix does not count toward the page limit. Each paper was found through Google Scholar, ACM Digital Library, or the publisher site. The "Checked against" column lists the source used to confirm each detail.

Ref	Title (short)	Venue, year	DOI / link	Checked against	Opened
[1]	Energy and AI	IEA report, 2025	iea.org/reports/energy-and-ai	IEA executive summary	☑
[2]	Making AI less "thirsty"	Commun. ACM, 2025	doi.org/10.1145/3724499	Author PDF of CACM article; vol./issue/pages on dblp	☑
[3]	Carbon-aware workflow scheduling	ICPP '25, 2025	doi.org/10.1145/3754598.3754673	Full text (arXiv:2507.08725); pages on KIT repository	☑
[4]	WaterWise	PPoPP '25, 2025	doi.org/10.1145/3710848.3710891	Full text, arXiv:2501.17944	☑
[5]	Energy-aware job shop, CP	Comput. Ind. Eng., 2025	doi.org/10.1016/j.cie.2025.111065	TU Dresden research portal abstract	☑
[6]	EV energy management, graph search	Energies, 2024	doi.org/10.3390/en17215278	Full text (MDPI)	☑
[7]	SHIELD	IGSC '23, 2023	doi.org/10.1145/3634769.3634810	Full text (Colorado State repository)	☑
[8]	Rule-based control of chillers	Energies, 2022	doi.org/10.3390/en15072419	RePEc abstract	☑
[9]	Discrete-time MDP policy	SMARTGREENS, 2023	doi.org/10.5220/0011846300003491	Full text (SciTePress)	☑
[10]	Data center-desalination coordination	Appl. Energy, 2026	doi.org/10.1016/j.apenergy.2026.127428	Abstract page	☑
[11]	DeepEE (parameterized DRL)	IEEE Trans. Serv. Comput., 2023	doi.org/10.1109/TSC.2022.3184835	BIT repository record (DOI, pages); dblp	☑
[12]	Carbon Explorer	ASPLOS '23, 2023	doi.org/10.1145/3575693.3575754	arXiv:2201.10036; PSU/ASU records	☑

All references are from 2022–2026, and seven of the eight reviewed studies use classical AI or constraint models. [1] is an agency report used only for background numbers.

Appendix B — AI Evaluation, Changes, and AI Use

(a) AI evaluation. The draft was pasted into a new AI chat and evaluated with Prompt 2B. Score: **8.2 / 10** (before the changes below). This is an AI-assisted self-assessment, not an instructor grade.

Criterion	Score	Strengths	Weaknesses	Recommended fix
Relevance and quality of papers (2)	1.8	Relevant, recent ACM, IEEE, and journal papers; cross-domain choices justified	Limited coverage of directly integrated water, cooling, and scheduling work	Justify why the selection covers the problem
Critical comparison (3)	2.3	Compares exactness, scalability, baselines, uncertainty, interpretability, validity	Numbers hard to compare across settings; closest methods not compared side by side	Compare WaterWise, SHIELD, and DeepEE on objectives and constraints
Research gaps (2)	1.7	Two clear gaps supported by DeepEE, WaterWise, Carbon Explorer	Gap not established beyond the reviewed literature	Keep qualified wording; state the missing integration precisely
Technical accuracy (1)	0.7	Good grasp of MILP, CP, A*, MDP, RL, heuristic search	CASSIA's heuristic not yet validated for its own cost model	Validate the heuristic against CASSIA's transition and cost model
IEEE references and organization (2)	1.7	Consistent numbering, DOIs, logical sections	Minor description details	Align descriptions with the original papers

(b) Changes made in response to the review and a Report 1 alignment check.

- Added the verified scale details of [3] (34 workflows; synthetic graphs of 200–30,000 vertices; nine Eager graphs up to 18,000; ILP only up to about 200 tasks).
- Compared WaterWise, SHIELD, and DeepEE side by side on objective, constraints, scheduling, cooling, water, and limitation, and stated the consequences for CASSIA.
- Removed the heat-pump assumption from CASSIA's heuristic, since Report 1 specifies none; CASSIA must now prove admissibility for its own cost model.
- Aligned Section 4 with Report 1: CSP job and cooling scheduling, rule-based cooling and water selection, A* heat allocation to heating loops, MD/FO, or thermal storage, and zero thermal-throttling for high-priority jobs.

(c) AI tools used. Claude (Anthropic) was used for paper discovery and summarization, drafting, and reference formatting; ChatGPT assisted with critical review, revisions, and the Prompt 2B evaluation; the team checked each paper against its source.